Phase 1 COMPLETED - Hashnode Clone MERN Platform

Status: 100% Complete
GitHub: https://github.com/Mainak-2006/hashnode-clone
Date: November 5, 2026

Phase 1 Summary

This phase established the foundational MERN stack setup for the developer blogging platform, including user authentication, database schema, and project scaffolding.

Completed Features

1. Project Scaffolding

- Created `server/` and `client/` directories
- Initialized Node.js project with `npm init -y`
- Installed core dependencies: `express`, `mongoose`, `jsonwebtoken`, `bcryptjs`, `cors`, `dotenv`
- Set up Vite React frontend setup

2. Backend Configuration

- `server/index.js` - Main Express entry point with MongoDB connection
- `server/.env.example` - Environment variable template
- `server/.gitignore` - Properly excludes `node_modules/`, `.vscode/`, `.env*`

3. User Model & Authentication

- `server/models/User.js` - Mongoose schema with:
  - Username (unique, trimmed, minlength 3)
  - Email (unique, validated format)
  - Password (minlength 6, hashed with bcryptjs)
  - Profile picture URL, bio (max 200 chars), timestamps
- Password hashing via `pre('save')` middleware

- `comparePassword` method for login validation

4. Authentication Controllers

- `server/controllers/authController.js`
  - `register` - Creates new user, returns success message
  - `login` - Verifies credentials, generates JWT token
  - `getCurrentUser` - Protected route placeholder
- JWT sign with 1-day expiration

5. Auth Routes

- `server/routes/auth.js` - Express router with:
  - `POST /register` - Public endpoint
  - `POST /login` - Public endpoint
  - `GET /me` - Protected endpoint (placeholder)

6. Blog Placeholder Routes

- `server/routes/blogs.js` - Basic CRUD structure
- Health check route `GET /`

7. Git History Management

- Removed `node_modules/` from git tracking
- Removed `.vscode/` from git tracking
- Force-pushed clean history to GitHub
- Updated `.gitignore` with proper entries

8. Documentation

- `README.md` - Comprehensive project documentation
- `PHASES.md` - 4-phase development roadmap (25%/50%/75%/100%)

Phase 1 File Structure

```
hashnode-clone/
├── .gitignore              # Git exclusions
├── PHASES.md               # Roadmap (4 phases)
├── README.md               # Project documentation
├── server/                 # Node.js/Express backend
```

```
│   ├── index.js            # Server entry point
│   ├── models/User.js      # Mongoose User schema
│   ├── controllers/authController.js  # Auth logic
│   ├── routes/auth.js      # Auth routes
│   ├── routes/blogs.js     # Blog placeholder routes
│   ├── .env.example        # Environment vars template
│   └── package.json        # Server dependencies
├── client/                 # React Vite frontend
│   ├── package.json        # Client dependencies
│   └── ... (pending Phase 2)
└── hashnode-clone.docx     # Original reference document
```

Technical Details

MongoDB Connection

```
mongoose.connect(process.env.MONGODB_URI || 'mongodb://localhost:27017/hashnode-clone')
  .then(() => console.log('   MongoDB connected'))
  .catch(err => {
    console.error('MongoDB connection error:', err);
    process.exit(1);
  });
```

User Schema Fields

Field	Type	Requirements
`username`	String	Required, unique, minlength: 3
`email`	String	Required, unique, valid email format
`password`	String	Required, minlength: 6
`profilePicture`	String	Default: ''
`profileBio`	String	Max: 200 characters
`createdAt`	Date	Default: Date.now

bcryptjs Configuration

- Salt rounds: 10 (industry standard)
- Hashing happens via `pre('save')` middleware
- Comparison via `user.comparePassword(candidatePassword)`

JWT Configuration

```
const token = jwt.sign(
  { userId: user._id },
```

```
  process.env.JWT_SECRET || 'default_secret',
  { expiresIn: '1d' }
);
```

How to Run Phase 1

Prerequisites

- Node.js (v14+)
- MongoDB (local or Atlas cloud)
- Git

Installation

```
# 1. Clone repository
git clone https://github.com/Mainak-2006/hashnode-clone.git
cd hashnode-clone

# 2. Install server dependencies
cd server
npm install

# 3. Install client dependencies
cd ../client
npm install

# 4. Set up environment variables
cd ../server
cp .env.example .env
# Edit .env with your MongoDB URI and JWT secret

# 5. Start the server
cd ../server
npm run dev   # or: node index.js
# Server runs on http://localhost:5000

# 6. Start the client (new terminal)
cd client
npm run dev   # Vite dev server
# App runs on http://localhost:5173
```

API Testing

```
# Register new user
curl -X POST http://localhost:5000/api/auth/register \
  -H "Content-Type: application/json" \
  -d '{"username":"testuser","email":"test@example.com","password":"password123"}'

# Login
curl -X POST http://localhost:5000/api/auth/login \
  -H "Content-Type: application/json" \
  -d '{"email":"test@example.com","password":"password123"}'
```

Phase Completion Metrics

Metric	Value
Phase Percentage	100%
Commits to GitHub	5
Files Created	12+
Lines of Code	~500+
Dependencies Installed	20+
API Endpoints	3 (register, login, me)
Models	1 (User)
Git History Cleanup	Complete

Phase 1 Checklist

- ☒ Project scaffolding (server + client folders)
- ☒ Express.js backend setup
- ☒ MongoDB connection with Mongoose
- ☒ User model with bcrypt password hashing
- ☒ JWT authentication (sign/verify)
- ☒ Registration API endpoint
- ☒ Login API endpoint
- ☒ Protected route placeholder
- ☒ Git initialization and first commit
- ☒ GitHub repository push
- ☒ Git history cleanup (node_modules, .vscode)
- ☒ .gitignore configuration
- ☒ README.md documentation
- ☒ PHASES.md roadmap
- ☒ Blog placeholder routes

Transition to Phase 2

Phase 2 Goals (50% completion):

- Blog post Mongoose schema creation
- Blog CRUD APIs with JWT protection
- Markdown rendering with `react-markdown`
- Syntax highlighting integration
- Draft saving & publishing functionality
- Blog listing with pagination

Next Steps:

1. Proceed to Phase 2 when ready
2. Install `react-markdown` and syntax highlighter in client
3. Create Blog model, controller, and routes
4. Build Markdown editor/preview components
5. Commit and push to GitHub

Author

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Inspired by: Hashnode

Phase 1 Complete — Ready for Phase 2: Blog CRUD & Markdown Features